First vent with centre O_1 $(12,0)$ and radius $r=10$ yields equation to first semicircle as

$$(x-12)^2+(y-0)^2=10^2$$

$$\Rightarrow x^2+y^2-24x+44=0,\ y>0.$$

Second vent with centre O_2 $(34,0)$ and radius $r=10$ yields equation to second vent as

$$(x-34)^2+y^2=10^2$$

$$\Rightarrow x^2+y^2-68x+1056=0,\ y>0.$$

EXERCISE 5.1

1. Obtain the equation of the circles with radius 5 cm and touching x-axis at the origin in general form.
2. Find the equation of the circle with centre $(2,-1)$ and passing through the point $(3,6)$ in standard form.
3. Find the equation of circles that touch both the axes and pass through $(-4,-2)$ in general form.
4. Find the equation of the circle with centre $(2,3)$ and passing through the intersection of the lines $3x-2y-1=0$ and $4x+y-27=0$.
5. Obtain the equation of the circle for which $(3,4)$ and $(2,-7)$ are the ends of a diameter.
6. Find the equation of the circle through the points $(1,0)$, $(-1,0)$, and $(0,1)$.
7. A circle of area 9π square units has two of its diameters along the lines $x+y=5$ and $x-y=1$. Find the equation of the circle.
8. If $y=2\sqrt{2}x+c$ is a tangent to the circle $x^2+y^2=16$, find the value of c.
9. Find the equation of the tangent and normal to the circle $x^2+y^2-6x+6y-8=0$ at $(2,2)$.
10. Determine whether the points $(-2,1),(0,0)$ and $(-4,-3)$ lie outside, on or inside the circle $x^2+y^2-5x+2y-5=0$.
11. Find centre and radius of the following circles.
    (i) $x^2+(y+2)^2=0$ (ii) $x^2+y^2+6x-4y+4=0$
    (iii) $x^2+y^2-x+2y-3=0$ (iv) $2x^2+2y^2-6x+4y+2=0$
12. If the equation $3x^2+(3-p)xy+qy^2-2px=8pq$ represents a circle, find p and q. Also determine the centre and radius of the circle.

G4X5GZ

5.3. Conics

Definition 5.2

A conic is the locus of a point which moves in a plane, so that its distance from a fixed point bears a constant ratio to its distance from a fixed line not containing the fixed point.

The fixed point is called **focus,** the fixed line is called **directrix** and the constant ratio is called **eccentricity**, which is denoted by ***e.***

(i) If this constant $e=1$ then the conic is called a parabola

(ii) If this constant $e<1$ then the conic is called a ellipse

(iii) If this constant $e>1$ then the conic is called a hyperbola

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